

14

Controlling morphosyntactic competition through phonology

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14.1 Introduction

This chapter discusses a case of suspended affixation in Turkish, i.e. the phenomenon in which certain affix(es) is affixed to the periphery of the coordination but interpreted for all coordinates. The main data discussed in this chapter involve stem allomorphy with 1SG and 2SG pronouns. The 1SG pronoun *ben* ‘I’ has a phonologically unexpected dative shape *ban-a* ‘I-DAT’ (instead of the expected **ben-e*). This chapter accounts for two unexpected patterns: (i) contra Guseva and Weisser (2018) and Erschler (2018), the suppletive form *ban-* that is a substring of *bana* is illicit under suspended affixation; and (ii) the personal pronouns are only acceptable in the first conjunct only with vowel harmonic joiners like *ve* (and) unlike *ya=da* (or). I propose a modification to the Lexicalization Algorithm (LA) (Starke 2020) in order to rule out a lexicalization as illicit on the grounds of phonotactic reasons. With the help of lexicalization movements and the algorithm provided in the LA, we achieve the necessary identity match for the ellipsis. The contribution of this chapter is to enrich the LA to enable phonology to have a say in the morphological computation.

14.2 Overview

This chapter aims to contribute to the discussion of how syntactic features are mapped to phonological realizations. Unlike the commonly employed LA, I argue that phonotactic factors, such as vowel harmony, can also influence which morphological form will be selected. To illustrate the interaction between phonotactics and the LA, and how it results in reranking of the morphological candidates, I will discuss suspended affixation data from Turkish. Consider examples (1) and (2) from Turkish and Digor Ossetic.

(1) Turkish (Kabak 2007)

Gid-er, gör-ür ve al-ır-ız.
 go-AOR see-AOR and buy-AOR-1PL

‘We go (there), see (it), and buy (it).’

Not Available: ‘S/he goes (there), s/he sees (it), and we buy (it).’

(2) Ossetic (Erschler 2012)

Alan ɐma Soslan-ɐj tarstɐn.
 Alan[NOM] and Soslan-ABL be.afraid.PST.1SG

‘I was afraid of Alan and Soslan.’

Sentence (1) provides an example where verbs are conjoined via the conjoiner *ve*, meaning ‘and.’ Only the final conjunct, *alırız*, carries the person marking *-ız*, yet all conjuncts are interpreted as if they were marked with the first-person plural marker. Even though both the first two conjuncts can be interpreted as 3rd person in the absence of overt agreement marking, as in ‘*S/he goes (there), s/he sees (it)*’, this reading is not available even in a context that might technically enable this reading. Similarly, in the Ossetic example, a case marker that appears at the right periphery of a coordinated nominal phrase takes scope over both conjuncts. Even though *Alan* surfaces in a bare form associated with the nominative case, it is interpreted as an ABL-marked nominal.

This phenomenon has been observed in various languages including Turkish (Kornfilt 1996, 2012; Kabak 2007; Broadwell 2008; Akkuş 2016; Atmaca 2022), Mari (Guseva and Weisser 2018), Ossetic (Erschler 2012), Iron (Erschler 2012), Eastern Armenian (Erschler 2012), Dagur (Gong 2021), Japanese (Yoon and Lee 2005), Korean (Yoon and Lee 2005), Nivkh (Gruzdeva 1998), and Hungarian (Trommer 2008). Figure 14.1 shows two prominent analyses of this phenomenon. Many previous analyses have noted the similarity of suspended affixation with the right-node-raising (RNR) phenomenon; these analyses interpret the suffixes in (1) and (2) as being attached to the coordination phrase (Broadwell 2008; Kornfilt 2012). A more recent postsyntactic deletion under recoverability analysis based on the case assignment in alternative questions was proposed by Erschler (2012, 2018) and Guseva and Weisser (2018), arguing that conjuncts are marked with the to-be-suspended affixes first, and coordinated later as in Figure 14.1B, as opposed to the RNR analysis as in Figure 14.1A.¹

¹ I leave the discussion of exactly why Ellipsis (deletion under identity) is preferred over the RNR analysis for another occasion since it goes beyond the scope of this chapter. See Gračanin-Yüksek (2016) and Erschler (2018).

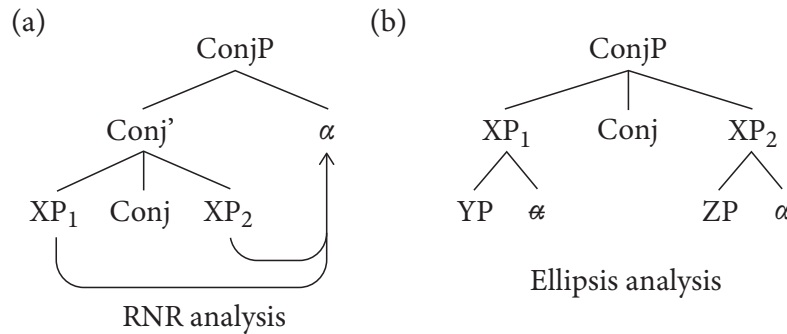


Figure 14.1 (a) RNR analysis. (b) Ellipsis analysis

Although this analysis has remained unchallenged, certain details are a topic of ongoing debate in morphology. One of the issues that has occupied morphologists concerns the details of the remnant in the ellipsis analysis. While some of the previous analyses argue that the remnant must be a word that can stand alone, namely, a morphological word (Kabak 2007; Erschler 2012), there is also evidence from Mari and Turkish that shows that suspended affixation does not need to leave behind a string that can stand alone (Guseva and Weisser 2018; Atmaca 2022). Example (3) shows an example from Mari. Speakers of Mari can leave the suppletive form *memna* behind even though the word itself is only available when the first-person plural personal pronoun *me* is in a syntactic position where the accusative case is assigned, and its suppletive form cannot be used alone anywhere. Similarly, the Mari nominative form *me* is unacceptable in suspended affixation contexts even though it is a substring of the accusative-marked pronoun (*pace* Erschler 2018).

(3) Mari (Guseva and Weisser 2018)

- a. Pörjeng memna(-m) da nunem už-eš.
 Man[NOM] 1PL.ACC and them.ACC see-3SG.PRS
- b. *Pörjeng me da nunem už-eš.
 Man[NOM] 1PL.NOM and them.ACC see-3SG.PRS
- ‘The man sees us and them.’

Even though Turkish does not need a morphological word in certain contexts involving sentence-level suspended affixation (Atmaca 2022), it is not freed from other constraints that are usually associated with morphological wordhood. Unlike Mari, Turkish speakers cannot leave suppletive forms behind as in (4). Previous papers that mention similar Turkish data argue that this is generally due to personal pronouns being generally ungrammatical in

suspended affixation contexts (Kabak 2007; Kornfilt 2012; Guseva and Weisser 2018).

(4) Turkish

*Ben ve sana mektup gel-miş.
1SG[NOM] and you.DAT letter arrive-EVID[3SG]
'Apparently, a letter arrived for me and you.'

However, the examples they used consist of conjoined phrases where both conjuncts are pronouns and have suppletive forms. The ungrammaticality is resolved when only the first conjunct is a suppletion-prone pronoun (5)(b). Given what has been attested in Mari and Ossetic (to be discussed more thoroughly in Section 14.3), one would expect the substring of the word *bana* to be grammatical in suspended affixation contexts, which is not the case in Turkish (5)(a).

(5) Turkish

- a. Ban*(-a) ve Olgun-a mektup gel-miş.
1SG.DAT and Olgun-DAT letter arrive-EVID[3SG]
b. Ben ve Olgun-a mektup gel-miş.
1SG[NOM] and Olgun-DAT letter arrive-EVID[3SG]
'Apparently, a letter arrived for me and Olgun.'

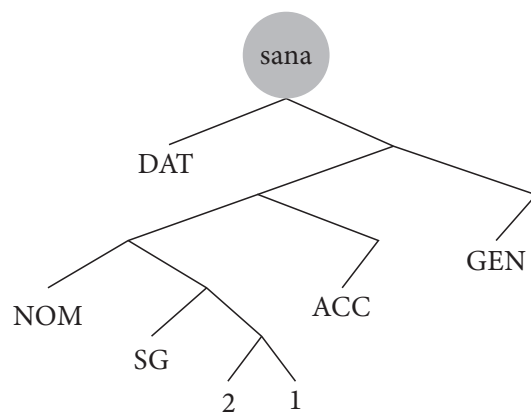
At first glance, these Turkish facts seem to be a counterexample to the postsyntactic deletion analysis. Yet, I will show that a closer look at the Turkish data will point us to a syntactic mismatch between the remnant and the suffix in the substring *ban* case (5)(a). However, the crash will be avoided in certain environments. In this chapter, I propose an analysis in which phonological processes are at play in the selection of allomorphy and the avoidance of the crash. I will argue that the ungrammatical lexicalization in (5)(a) will be re-evaluated in the presence of the vowel harmonic restrictions imposed by the joiner *ve*, which ends up being in the same phonological word as the first-person pronoun. In addition, I will argue that pronouns like *sana* and *bana* have complex structures that do not allow the decomposition of *-a* at all, resulting in an identity mismatch which explains why it is impossible to have either as a second conjunct as in (4). Identity match between the first and the second conjuncts will only be available when *ve* forces a backtrack operation to have the decomposition of *sen+A* in the first conjunct, as proposed in Türk and Caha (2022). However, since there will be no phonotactic

constraints imposed on the second conjunct, the only grammatical lexicalizations will be the ones where only the first conjuncts are pronouns. In other words, there is a mismatch between what needs to be deleted to get from *bana* to *ben* and the decomposition of *san* and *-a*, but that is not the case with other non-suppletive nouns like *Olgun-a*.²

This proposal is building on previous work from Svenonius (2012) and Bye and Svenonius (2012), who proposed a similar model in which the lexical insertion is divided into two parts: (i) phonology-free syntax and (ii) syntax-free phonology. One of the main pieces of evidence they present is the French preposition–determiner fusion. French determiners have three basic forms: [lə], [la], and [l], for masculine, feminine, and vowel-initial words, respectively. In the context of certain prepositions like *à*, [a], we find [ala] and [al], but not the expected [alə]. Similarly, in the context of *de* [də], we find [dəla] and [dəl], but not the expected [dələ]. Instead, prepositions and determiners fuse when the noun starts with a consonant, resulting in [o] and [dy] for the expected [alə] and [dələ], respectively.

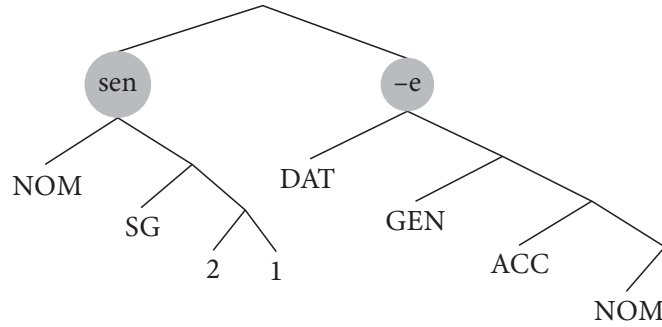
Similar to what I am proposing in this chapter, they argue that to be able to solve this problem, syntax and phonology have to be interacting with each other. Following this enrichment from Svenonius (2012) and Bye and Svenonius (2012), I propose a similar explanation for the Turkish suspended affixation data. In addition to previous work in Nanosyntax and case analysis of Turkish, I propose the following functional sequence and lexical items for the Turkish DAT case paradigm as in (6)(a) and (6)(b) (Caha 2009; Starke 2017; Türk and Caha 2022).

(6) a. Stored Lexical Unit for *sana*



² Even though a version of this was introduced in Kornfilt (2012), the model presented there was not equipped to handle both (4) and (5), for the model presented there only eliminated the ungrammatical forms but did not have the generative power to account for (5)(b).

b. Possible decomposed unit



Note that (6)(b) is the usual dative case that is found with common nouns as proposed by Türk and Caha (2022), and it is the result of a forced backtracking due to the phonological interference by the conjainer *ve* as in (5)(b).

14.3 Suspended affixation in Turkish

Suspended affixation in Turkish can surface in many different environments, including nominals, verbs, derivational, and inflectional contexts (Göksel and Kerslake 2005; Akkuş 2016). The acceptability of the suspended affixation is not affected by any type of morphophonological process (Kabak 2007), or the syntactic complexity of the elements taking part in suspended affixation.

What seems to matter is the syntactic and feature identity of the elements omitted. Even though PL, POSS, and ACC can be suspended either by themselves or in various combinations with each other ((7)(a), (7)(b), (7)(d)), there is a limitation on the environments in which PL and POSS can be separated. POSS cannot be suspended alone when there is a PL marker as in (7)(c). Note that this is not due to simply not being able to have PL as a final suffix (7)(e) (Kabak 2007).

(7) Turkish (Kabak 2007)

- a. kedi-ler-im-i ve köpek-ler-im-i
 cat-PL-POSS.1SG-ACC and dog-PL-POSS.1SG-ACC
- b. kedi-ler-im ve köpek-ler-im-i
 cat-PL-POSS.1SG and dog-PL-POSS.1SG-ACC
- c. *kedi-ler ve köpek-ler-im-i
 cat-PL and dog-PL-POSS.1SG-ACC
- d. kedi ve köpek-ler-im-i
 cat and dog-PL-POSS.1SG-ACC
 'my cats_{ACC} and my dogs_{ACC}'

- e. kedi-ler ve köpek-ler-de
 cat-PL and dog-PL-LOC
 ‘cats_{LOC} and dogs_{LOC}’

However, the same pattern of inseparability is not observed with a certain set of nouns with a collective reading (8).

(8) Turkish (Kabak 2007)

- a. asker-ler ve komutan-lar-ımız-ı
 soldier-PL and commander-PL-POSS.1PL-ACC
 ‘our soldiers_{ACC} and our commanders_{ACC}’
 b. avukat-lar ve danışan-lar-ınız-ı
 lawyer-PL and consultant-PL-POSS.2SG-ACC
 ‘your lawyer_{ACC} and your consultants_{ACC}’

Even though there is no established analysis of why (7)(c) is ungrammatical but (8)(a)–(b) is not, I argue that the PL in (8) that attaches to possibly collective nouns and the PL in (7) that attaches to count nouns are different and have different bracketing as in (9).

- (9) a. Bracketing for (7): [NP [N_{count} cat] [PossP [PL] [POSS]]]
 b. Bracketing for (8): [NP [NP [N_{collective} soldier] PL] POSS]

Another important data point comes from the derivational morphology and optionality of the suspended reading. There are also cases in which conjoining a bare NP and an NP with a derivational morpheme result in sentences with ambiguous meanings (10).

(10) Turkish (Bozşahin 2007)

- tuz ve limon-luk
 salt and lemon-container
 a. ‘salt shaker and lemon squeezer’ (SA)
 b. ‘salt and lemon squeezer’ (No SA)

Kornfilt (2012) notes that the ordering of the elements matters in certain elements like (10). When *tuz*, which can be used to refer to the item saltshaker by itself as in (11)(a), comes after *limon*, which cannot refer to the lemon squeezer as in (11)(b), the container suffix *-luk* does not create ambiguous readings as in (11)(c).

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(11) Turkish (Kornfilt 2012)

- a. Tuz-u uzat.
salt-ACC pass.IMP
'Pass the salt (shaker).'
- b. Kokteyl için limon-u kullan-ma.
cocktail for lemon-ACC use. IMP-NEG
'Do not use the lemon for the cocktail.'
* 'Do not use the lemon squeezer for the cocktail.'
- c. limon ve tuz-luk
lemon and salt-container
 - i. * 'lemon squeezer and salt shaker' (SA)
 - ii. 'lemon and salt shaker' (No SA)

Akkuş (2016) demonstrated that this is due to the slightly different complex structure of the nominals. He provides a counter explanation to this asymmetry, arguing that the reason for not having both readings in (12) is because *lemon* is countable while *salt* is a mass noun, and two syntactically distinct elements. When both nouns are uncountable, for example, replacing *lemon* with (*black*) *pepper*, the problem discussed in Kornfilt (2012) is avoided and ambiguity is available.

(12) Turkish (Akkuş 2016)

- biber ve tuz-luk
pepper and salt-container
- a. 'pepper mill and salt shaker' (SA)
- b. 'pepper and salt shaker' (No SA)

(13) Turkish (Akkuş 2016)

- tuzve biber-lik
saltand pepper-container
- a. 'saltshaker and pepper mill' (SA)
- b. 'salt and pepper mill' (No SA)

Both observations involving PL and POSS or the derivational morpheme *-lik* suggest that the asymmetries found in Turkish suspended affixation are due to the minute structure of the nominals and that the syntactic identity of deletion and the remnant matter.

14.4 Ban on non-wordhood and root allomorphy

Even though the ellipsis analysis provides a uniform analysis for Mari, Turkish, and Ossetic, it is far from a complete explanation of the facts of these specific languages. Erschler (2018) provides seven additional descriptive properties that do not directly follow from the ellipsis analysis. Similarly, Guseva and Weisser (2018) provide a rule-ordering mechanism for Turkish that differs from Mari to capture differences between languages.

One important difference between these languages that has been discussed frequently is the form-wise characteristics of the remnant of the suspended affixation, the first conjunct. Both Erschler (2018) and Kabak (2007) argue that the remnant must be an independent ‘stand-alone’ word, meaning that what is left behind should be such that it could be freely used in other contexts other than suspended affixation.

14.4.1 Ossetic and Mari

Consider the simplified version of Ossetic case paradigm with the second-person pronouns given in (14).

(14) Ossetic second-person singular pronouns

Case	Form
NOM	<i>du</i>
ACC/GEN	<i>dew</i>
DAT	<i>dew-en</i>
ABL	<i>dew-ej</i>

As noted by Erschler (2018), all non-nominative-marked forms of the second singular pronouns are parasitic on the accusative/genitive case. The nominative case form does not surface in any of the cases, in any of the singular pronouns.

When the ablative-marked second-person singular pronoun is uttered as the first conjunct in the environment of suspended affixation, the ablative-marked pronoun does not surface as *du*, but instead surfaces as *dew* as in (15).

(15) Ossetic (Erschler 2018)

<i>dew</i> /* <i>du</i>	<i>ema</i>	<i>Alan-ej</i>	<i>tersun.</i>
you.ACC/*NOM	and	Alan-ABL	fear.PRS.1SG
‘I am afraid of Alan and you.’			

However, *kereḏze*, meaning ‘each other’ cannot be left alone in the context of suspended affixation as in (16), even though it is frequently used with case suffixes in Ossetic, suggesting high decomposability. The main reason behind this mismatch is that *kereḏze* never occurs in a non-case-marked form, so the word does not exist by itself even though it is easily decomposable.

(16) Ossetic (Erschler 2018)

- a. *nɐ=duwɐ tikiš-i kereḏze ɐmɐ nɐ=kuj-ɐj
 our=two cat-ACC each.other and our=dog-ABL
 tɐrsuncɐ.
 fear.PRS.3PL
- b. nɐ=duwɐ tikiš-i kereḏze-ɐj ɐmɐ nɐ=kuj-ɐj
 our=two cat-ACC each.other-ABL and our=dog-ABL
 tɐrsuncɐ.
 fear.PRS.3PL
- ‘Our two cats are afraid of each other and of our dog.’

Erschler (2018) explains this behavior in terms of constraints on the suspended affixation. Constraint (D) says that ‘remnants [...] must be substrings of the respective full forms,’ inhibiting the surface form *du*. Another important constraint is the Constraint (E), which is the ‘stand-alone’ condition, inhibiting the non-affixed *kereḏze* from surfacing.

These constraints, however, are not easily transferable to other languages with suspended affixation. This arbitrariness raises the question of which part of the grammar these constraints reside in, and which modules they can speak to. For Erschler, suspended affixation is a process of phonological deletion and happens at the PF interface. Guseva and Weisser (2018) provide an even more explicit position. They argue that there are multiple places in PF that suspended affixation can occur in their attempt to generalize their analysis to Turkish suspended affixation.

The constraints that Erschler (2018) proposed are not an issue for Meadow Mari. Guseva and Weisser (2018) show that suppletive forms of words that do not surface by themselves in any other context can be legitimate candidates for remnant in suspended affixation contexts. For example, *memnam* (1PL.ACC) is a suppletive form of *me* (1PL.NOM), and its final sound *-m* is a shared ending with certain pronominal forms marked with the accusative case, making a case for decomposability. However, *memna* by itself is not available in any context; it is only used if there is any marking or ending coming right after it. Unlike *dew* in Ossetic, it is grammatical for Mari speakers to leave it as a remnant in suspended

affixation contexts; example (17)(b) should be ungrammatical if Mari were to behave like Ossetic, yet it is not.

(17) *Mari* (Guseva and Weisser 2018)

- a. Pörjeng memnam da nunem už-eš.
 Man.NOM us.ACC and them.ACC see-3SG-PRS
- a. Pörjeng memna da nunem už-eš.
 Man.NOM us.??? and them.ACC see-3SG-PRS
- ‘The man sees us and them.’

14.4.2 Turkish

Turkish, on the other hand, provides a different story from both languages. The issue of suppletion in Turkish suspended affixation has not received much attention. Its characteristics are more similar to Ossetic than Mari in two important aspects. However, the analysis of Ossetic cannot be extended to the Turkish data, nor can the already existing analyses, two-level PF (Guseva and Weisser 2018) or Phonological Cohesion (Kabak 2007) cover the entire data.

First, unlike Mari and similar to Ossetic, Turkish sometimes does not allow suspended affixation when both conjuncts are pronouns (Kabak 2007; Guseva and Weisser 2018). The example in (18) is ungrammatical, even though there is no reason that is easily derivable from the previously mentioned patterns in the literature.

(18) Turkish

- *İlk önce sen ve bana bak-tı.
 first you and I.DAT look-PST[3SG]
- Intended: ‘S/he first looked at you and me.’

However, analyzing this ungrammaticality as due to both conjuncts being pronouns would be a mistake. The ungrammaticality persists in sentences in which only the second conjunct is a pronoun. The problem, at least in Turkish, is not about having two pronouns as conjuncts, but specifically when the second conjunct is a pronoun as in (19).

(19) Turkish

- *İlk önce Olgun ve sana bak-tı.
 first Olgun and you.DAT look-PST[3SG]
- Intended: ‘S/he first looked at Olgun and you.’

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The ungrammaticality is not due to the animacy hierarchy, since using any other pronoun or having NPs that denote lower elements in the animacy hierarchy in the first conjunct does not change the grammaticality of the sentence, as in (20):

(20) Turkish

- a. *Olgun ve bana/biz-e/siz-e/on-a/onlar-a bak-tı.
 Olgun ve I.DAT/we-DAT/you-DAT/(s)he-DAT/they-DAT look-PST[3SG]
 Intended: ‘He looked at Olgun and me/us/you/him/her/them.’
- b. *Kedi ve bana/biz-e/siz-e/on-a/onlar-a bak-tı.
 cat ve I.DAT/we-DAT/you-DAT/(s)he-DAT/they-DAT look-PST[3SG]
 Intended: ‘He looked at the cat and me/us/you/him/her/them.’

However, the ungrammaticality is resolved when the second conjunct is not a pronoun as in (21). These examples with a non-pronoun second conjunct that were not discussed before undermine Kabak’s (2007) proposal about phonological cohesion due to suppletion.

(21) Turkish

- İlk önce sen ve Olgun-a bak-tı.
 first you and Olgun-DAT look-PST[3SG]
 Intended: ‘S/he first looked at Olgun and you.’

Moreover, a simple PF-deletion story with a substring constraint, as in Erschler (2018), would deem sentences like (22) grammatical. However, this is not the case. The substring *ban* is not an appropriate remnant in Turkish. This ungrammaticality alone is easy to explain with the Ossetic constraints, since *ban* is not a stand-alone word in Turkish, meaning that even though it is decomposable into *ban-* and *-a*, it does not occur in any other context without any suffixes.

(22) Turkish

- a. *Ban ve Okan-a mektup gel-miş.
 I.??? and Okan-DAT letter arrive-EVID[3SG]
- b. Bana ve Okan-a mektup gel-miş.
 I.DAT and Okan-DAT letter arrive-EVID[3SG]
 ‘A letter for me and Okan has arrived.’

However, unlike Ossetic, Turkish can leave non-substrings behind as in (21). However, this is also only available in the first conjunct. The solution of this mismatch is not straightforward for Erschler’s (2018) analysis of Ossetic. The interaction between conjunct order and the suppletion tells us that there is more to suspended affixation than just the phonological deletion, which is also supported by independent arguments by Caha (2019).

Even though many examples here follow most of the tenets of Erschler's (2018) analysis, the grammaticality of the non-substring *Ben* and the ungrammaticality of the substring *Ban* challenge his property of Constraint D. It is also not clear how Guseva and Weisser's (2018) analysis can capture the asymmetry between the first and the second conjunct. In their paper, they only report on ungrammatical suspended affixations in which the second conjunct is a pronoun. This enables them to say that in Turkish, unlike Mari, vowel harmony and similar phonological processes precede the suspended affixation, making sentences like (19) ungrammatical. However, for their model to work, we must speculate that the vowel harmony with common nouns like *Melek* (23) and the one with pronouns (19) occurs in different PF levels.

What is more puzzling is that with other case-markings the use of pronouns, even in both conjuncts, is completely grammatical as in (23) and (24) reported by Kabak (2007).

(23) Turkish

Sen ve Melek-i gör-müş.
 You and Melek-ACC see-EVID[3SG]
 'S/he apparently saw you and Melek.'

(24) Turkish (Kabak 2007)

Ben ve sen-den nefret ed-iyor.
 I and you-ABL hate AUX-PROG[3SG]
 'S/he hates me and you.'

These facts show that Turkish suspended affixation cannot be simply a PF operation or rule ordering without any reference to the internal structure of the pronouns, case, and common nouns. Thus, I propose an account using the Nanosyntax paradigm (Starke 2009), specifically because of the ability to keep track of the algorithmic history within this framework, which will be important for us to capture identity match.

14.5 Analysis

Following Erschler (2018), I assume a deletion analysis, in which structures are built first, coordinated, and then suspended suffixes are omitted later via deletion. I also assume that the pronouns involve three features: [speaker], [participant], and [person], which stand in a containment relation (see Béjar 2003; Starke 2013; Vanden Wyngaerd 2018). For convenience, I represent these features as 1, 2, and 3, respectively.

For the rest of the case system, I assume the proposal that was done in Türk and Caha (2022), following the case containment proposal by Caha (2009). Lastly, I am

using a realizational morphology account in which only phrasal nodes are lexicalized and in which lexicalization happens at every merge, following the algorithm proposed in Starke (2018).

In this section, I will first demonstrate what Türk and Caha (2022) proposed for the Turkish case system and will extend their analysis to pronouns with suppletive morphology. Later, I will demonstrate the behavior of pronouns in conjunction and suspended affixation environments. Lastly, I will show how the derivational history will help us decompose a suppletive unit and how it will save sentences like (24) from a derivational crash, but not the ones like (19).

14.5.1 Turkish case system

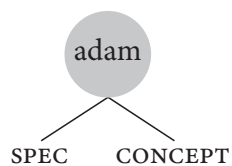
Türk and Caha (2022) proposed a Nanosyntactic analysis of the Turkish case system with the lexical items in (25).

(25) Lexical items proposed by Türk and Caha (2022)

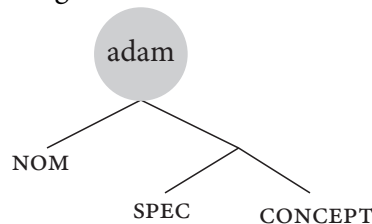
<i>adam</i>	$\Leftrightarrow$ [NOM (K1) [SPEC [CONCEPT]]]
<i>-ı</i>	$\Leftrightarrow$ [ACC (K2)]
<i>-n</i>	$\Leftrightarrow$ [GEN (K3)]
$\emptyset$	$\Leftrightarrow$ [GEN (K3) [ACC (K2) [NOM (K1)]]]
<i>-a</i>	$\Leftrightarrow$ [DAT (K4) [GEN (K3) [ACC (K2) [NOM (K1)]]]]

Their main aim was to model the containment relation between accusative and genitive case, as well as the zero-marking of ACC and GEN case in nonspecific nouns. To this end, they specified nouns with SPEC and NOM features. When a noun is specific, up until the nominative case, everything will be lexicalized with the noun itself, and additional cases will be lexicalized by their own lexical items. When the noun is not specific, only CONCEPT will be lexicalized by the noun, and the cases up until DAT will be lexicalized by zero morphology. For our purposes, we only need to look at the specific cases. Let's go over how DAT-marked nouns are modeled under the Nanosyntax framework. *Adam* is lexically specified for CONCEPT, NOM (K1), and SPEC and will lexicalize these features. The structures are assembled cyclically in (26). Just merging two terminal nodes will be enough for the LA to find a match.

(26) a. Merge SPEC

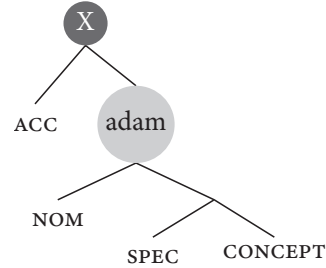
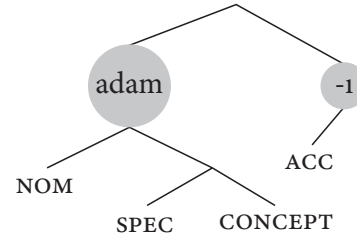


b. Merge NOM



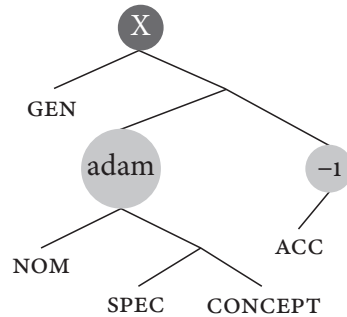
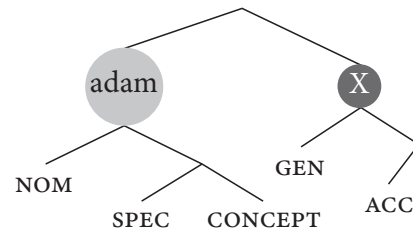
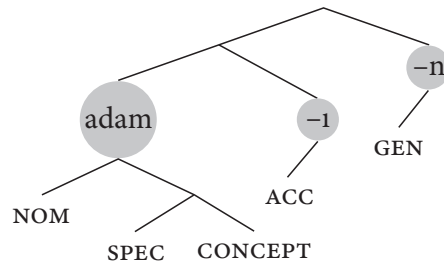
When we add the ACC (K2) feature, it will not be able to be lexicalized with a single lexical item. When a lexical match is not found, lexicalization-related left branch movements will kick in as specified by the LA (Starke 2018). Since the structure in (27a) lacks a Complex Left Branch, *move-sister* will occur, and ACC will be lexicalized in its own phrase.

(27) a. Merge ACC

b. *move-sister*

Upon merging, GEN (K3) will attempt to lexicalize as a single lexical item and will fail again due to the specifications of the lexical items in (25). First, *move-specifier* will be employed as in (28)(b). When *move-specifier* fails, *move-sister* will apply and give us the structure and lexicalization in (28c).

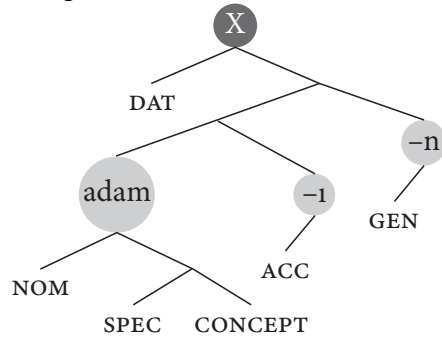
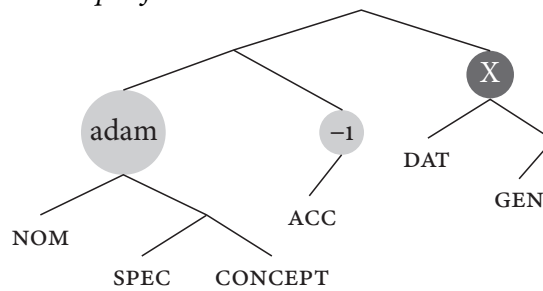
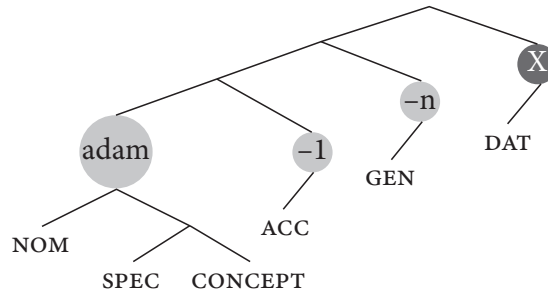
(28) a. Merge GEN

b. *move-specifier*c. *move-sister*

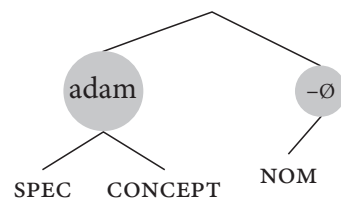
Finally, when DAT is merged, none of the lexicalization-driven movements will give us any licit lexicalizations as in (29).

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(29) a. Merge DAT

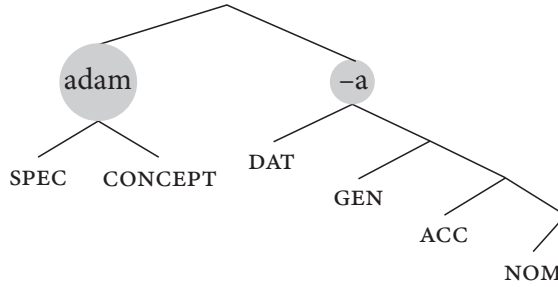
b. *move-specifier*c. *move-sister*

Thus, backtracking will occur. Backtracking is an operation where everything is undone until the most recent successful lexicalization, and do the next available step in the LA in the hopes of lexicalizing our current merged node, in this case it is DAT. In the lexicalization of both ACC and GEN, we have used the last step before backtracking as well. Therefore, we go back to the lexicalization of NOM and instead of staying within the root node as in (26)(b), *move-sister* is applied and matches the NOM phrase with the nonspecific ending as in (30).

(30) *move-sister* for NOM

From this point forward, we will merge features and repeatedly try to lexicalize the structure. At every merge, it will fail, but the left-branch *move-specifier* movement will rescue the structure, eventually resulting in the structure in (31).

(31) Backtracked DAT



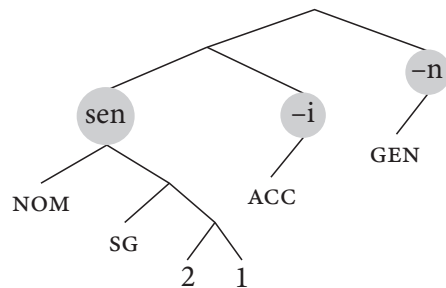
14.5.2 Extending Türk and Caha (2022) to pronouns

I propose the following lexical items in (32) for Turkish pronouns that exhibit suppletive morphology.

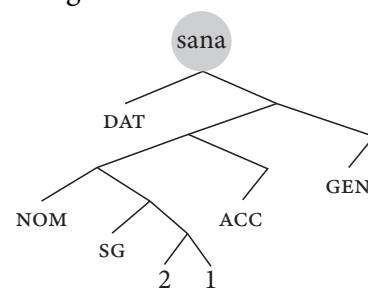
- (32) *sen* ⇔ [NOM (K1) [SG [2 [1]]]]
ben ⇔ [NOM (K1) [SG [1]]]
benim ⇔ [GEN (K3) [[NOM (K1) [SG [1]]] [ACC (K2)]]]
sana ⇔ [DAT (K4) [[[NOM (K1) [SG [2 [1]]]] [ACC (K2)]] [GEN (K3)]]]
bana ⇔ [DAT (K4) [GEN (K3) [[NOM (K1) [SG [1]]] [ACC (K2)]]]]

With the lexical items in (32) and (25), the structure will closely follow the ones in section 14.5.1 for the second-person pronoun *sen* until merging the DAT node as in (33)(a). When the DAT is merged, instead of triggering a *backtrack*, there will be a lexical match with *sana* as in (33)(b).

(33) a. Lexicalization until DAT

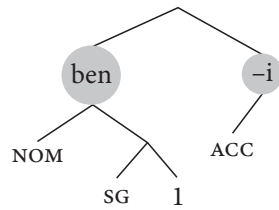


b. Merge DAT and match

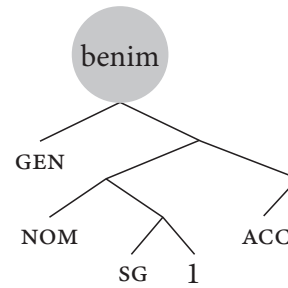


The process will be slightly different for the first-person pronoun *ben*, which exhibits a different exponence when marked with GEN *benim*, instead of the expected *ben-i-n*. While the structure will follow Türk and Caha (2022) until the merge of GEN as in (34)(a), GEN will not trigger the *move-specifier* or *move-sister* components of the algorithm. Instead, we will have a lexical match overwriting the previous lexicalizations (34)(b). Upon the merge of DAT, the same thing will happen (34)(c) given our lexical items in (32).

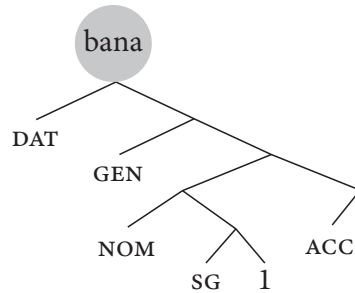
(34) a. Lexicalization until GEN



b. Merge GEN and match



c. Merge DAT and match



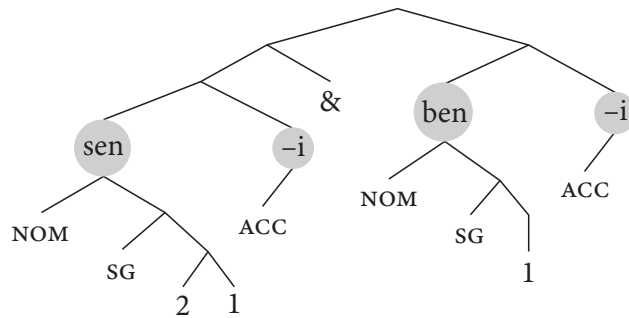
14.5.3 Suspended affixation and identity match

Given the lexical items and the syntactic complexity of pronouns and nominals, let us see how they behave in a suspended affixation context. Remember that for suspended affixation, we assume that it is an ellipsis-like process that requires an identity match between deleted nodes.

14.5.3.1 ACC

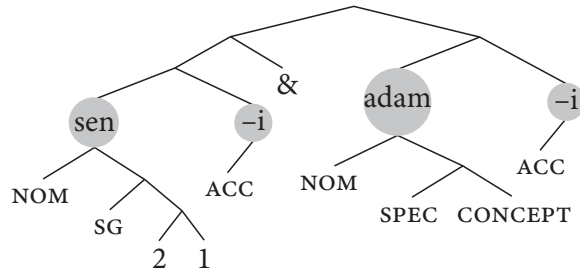
As we have shown before and reported in Kabak (2007), suspended affixation with two pronouns marked with ACC is acceptable. The structure is shown in (35). One can easily target the node that dominates ACC.

(35)



Since the structures of common nouns are very similar to that of pronouns when they are ACC-marked, suspended affixation is licit as both conjuncts have suffixal nodes that match in identity as in (36).

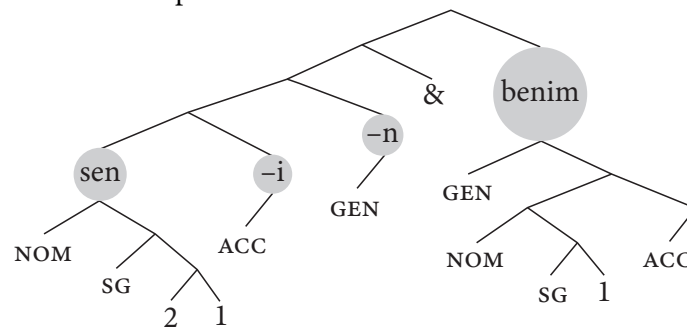
(36)



14.5.3.2 GEN

Due to the different complexity of GEN-marked first- and second-person pronouns (37)(a), we cannot delete the nodes *-i* and *-n*. This prediction is borne out given that sentences like (37)(b) are ungrammatical.

(37) a. GEN-marked pronouns



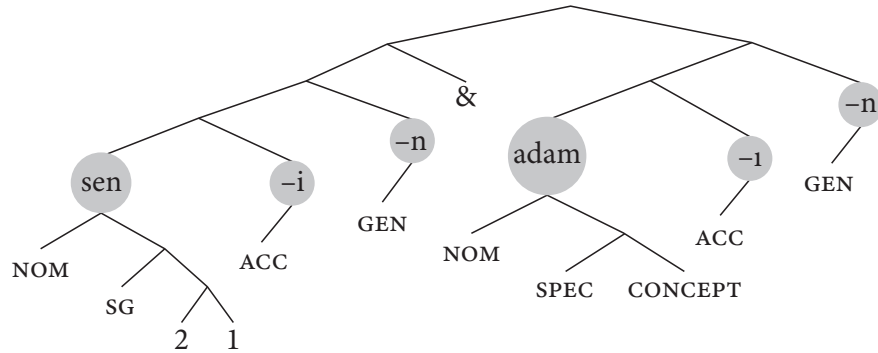
- b. Sen*(-in) ve ben-im araba sat-ıl-mış.
 You-GEN and I-GEN.1SG car sell-PASS-EVID[3SG]
 'Apparently, your and my car got sold.'

What this structure predicts is that the GEN marking on the second-person pronoun should be suspended with a second conjunct common noun (38)(a); however, this should not be possible with the first-person pronoun (38)(b). These predictions as well are borne out, and the structures are provided in (39).

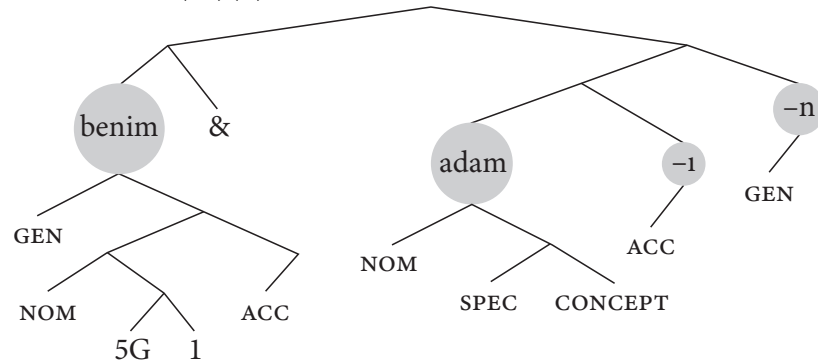
(38) Turkish

- a. Sen(-in) ve adam-in araba-sı sat-ıl-mış.
 You-GEN and man-GEN car-POSS sell-PASS-EVID[3SG]
 ‘Apparently, your and the man’s car got sold.’
- b. Ben*(-im) ve adam-in araba-sı sat-ıl-mış.
 I-GEN.1SG and man-GEN car-POSS sell-PASS-EVID[3SG]
 ‘Apparently, my and the man’s car got sold.’

(39) a. Structure for (38)(a)



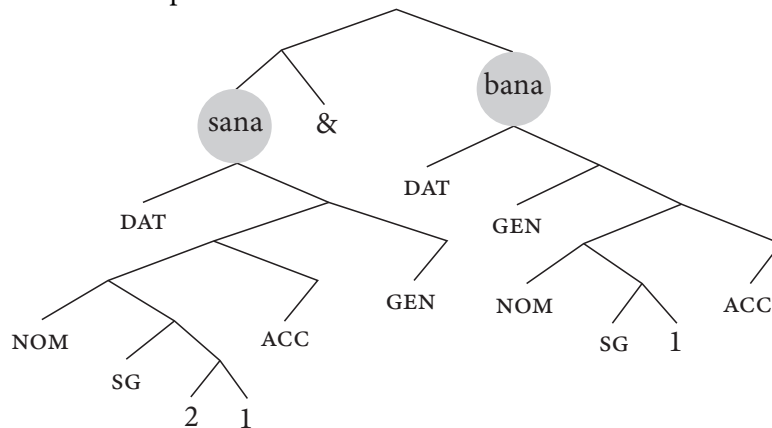
b. Structure for (38)(b)



14.5.3.3 DAT

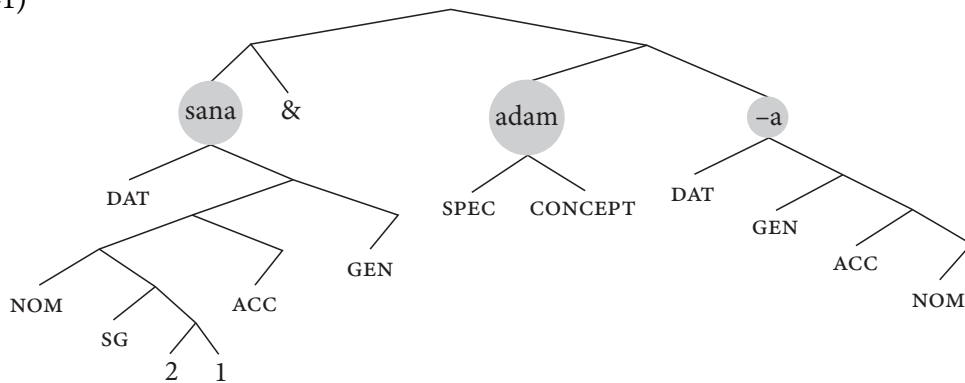
Lastly, due to same reasons with GEN, suspended affixation with two DAT-marked pronouns will be ungrammatical as in (18). The structure is presented in (40). Note that there is no single node that can be targeted since both *sana* and *bana* are single lexical items.

(40) DAT-marked pronouns



Given our lexical items above and the conjunction structure in (41) where we cannot target a single DAT node between two conjuncts, we should expect the same ungrammaticality with pairs of pronouns and common nouns. However, this is not the case as previously shown in (41).

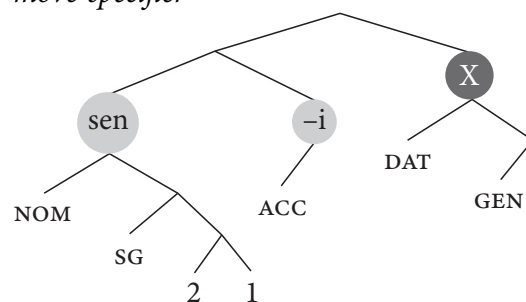
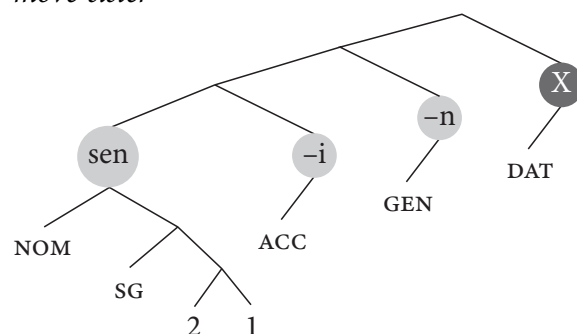
(41)



What is interesting about (41) is that, I propose, the attempt of deletion applied to the string, and not the structure, will create a phonological world between the conjunctive *ve* and the first conjunct, which in turn triggers a repair mechanism. This initiation of the repair mechanism is the phonology's contribution to the morphosyntactic computation. It will force the *sana* part of the structure to seem like it failed to lexicalize.

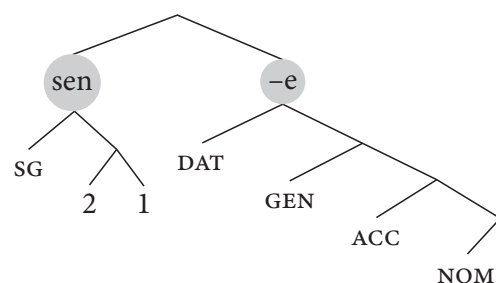
Remember, in our lexicalization process, after we merge DAT with the second-person pronouns, we immediately found a lexical match. However, now due to phonological reasons, that match will be ruled out as illicit, and we will apply *move-specifier* and *move-sister* steps, neither of which will give rise to a successful lexicalization as in (42)(a) and (42)(b).

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(42) a. *move-specifier*b. *move-sister*

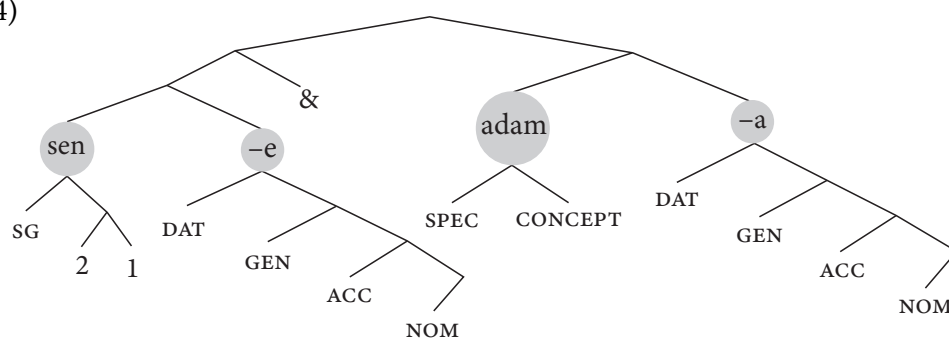
This will force us to apply the *backtrack* step to lexicalize DAT as we did with common nouns in (31). The resulting structure will look like the one in (43).

(43)



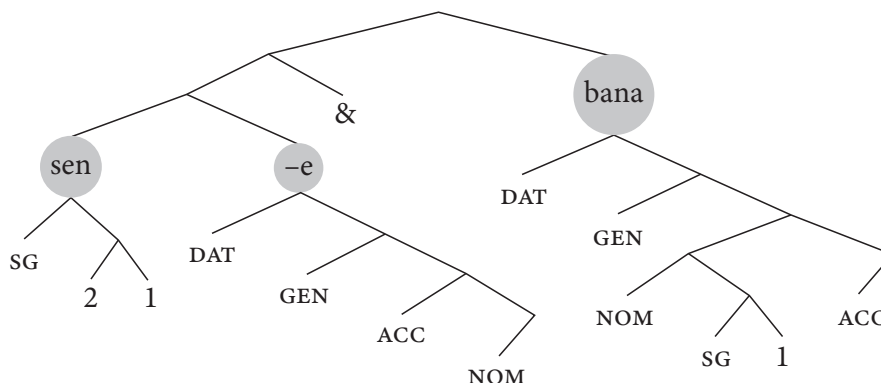
This structure, in the context of suspended affixation (44), will enable us to suspend DAT when the pronoun is the first conjunct.

(44)



Even if the same mechanism were triggered for the two pronouns cases (45), we would still not be able to suspend DAT. This is because the *bana* part of the conjunction will not be able to lexicalize as *ben-e* due to no phonological constraint on the second conjunct.

(45)



14.6 Ungrammatical nonharmonic conjoiners

One issue I have not discussed yet is the behavior of conjoiners like *ya da*, meaning ‘or,’ that are not harmonic with remnants like *ben* or *sen*. One possibility is that since these conjoiners are not problematic with the substrings *ban* or *san* in terms of vowel harmony, which would make (46) grammatical. However, this is not the case, as these sentences are not unacceptable and the presence of a conjoiner with a vowel that has a [+back] vowel as its initial vowel is not enough to make the non-stand-alone word *ban* appear as a remnant.

- (46) *Ban/San ya da Olgun-a mektup gel-miş.
 I.???/you.??? or Olgun-DAT letter arrive-EVID
 Intended: ‘A letter for me/you and Olgun has arrived.’

Another possibility is that suspended affixation is not possible with nonharmonic conjoiners. The sentence in (47) confirms this prediction. When the first conjunct and the conjoiner have mismatching vowel qualities, the conjoiner cannot initiate a *backtrack* operation since the other candidate *ban* is ungrammatical due to other reasons, namely the ban on nonwords.

- (47) *Ben/Sen ya da Olgun-a mektup gel-miş.
 I/you or Olgun-DAT letter arrive-EVID
 Intended: ‘A letter for me/you or Olgun has arrived.’

To test the possibility of personal variance, I conducted a speeded acceptability judgment task with sentences like (47) and their non-suspended versions. I hypothesized that if phonological processes influence the morphological

constituency, non-harmonizing conjoiners will significantly decrease the acceptability of sentences like (47), compared to harmonizing conjoiners.

14.6.1 Participants

All participants (N=170) were native Turkish speakers (age range: 18–59, M = 21). The experiment was carried out following the principles of the Declaration of Helsinki and the regulations concerning research ethics at Boğaziçi University. All participants provided their informed consent before participating and their identities were anonymized.

14.6.2 Materials

Participants were asked to judge forty experimental sentences as in (48) featuring manipulations of suspended affixation and conjoiner. All experimental items started with a personal pronoun that is susceptible to root allomorphy (*ben* or *sen*) in either its bare or marked form. Pronouns were followed by a conjoiner that is either harmonic with the bare form of the pronoun (*ve*) or not (*ya da*). The distance between the case-marked elements and the case assigner verb phrase was kept minimal, only intervened by a pseudo-incorporated subject or object. Experimental sentences were distributed among four different lists following a Latin-square design. In addition to experimental items, participants saw eighty filler items, half of which were ungrammatical.

(48) a. *Nonharmonic—Suspended affixation

Ben ya da Olgun-a mektup gel-miş.
I or Olgun-DAT letter arrive-EVID
Intended: ‘A letter for me or Olgun has arrived.’

b. Harmonic—Suspended Affixation

Ben ve Olgun-a mektup gel-miş.
I and Olgun-DAT letter arrive-EVID
‘A letter for me and Olgun has arrived.’

c. Nonharmonic—No Suspended Affixation

Bana ya da Olgun-a mektup gel-miş.
I or Olgun-DAT letter arrive-EVID
‘A letter for me or Olgun has arrived.’

d. Harmonic—No Suspended Affixation

Bana ve Olgun-a mektup gel-miş.
I and Olgun-DAT letter arrive-EVID
‘A letter for me and Olgun has arrived.’

14.6.3 Procedure

The experiment was run online, using the web-based platform IbexFarm (Drummond 2013). Each experimental session took approximately 40 minutes. Participants gave informed consent to participate in the experiment. They then proceeded to read the instructions and were given nine practice trials.

Each trial began with a blank screen for 600 ms, followed by a word-by-word Rapid-Serial-Visual-Presentation, and then an acceptability question. Sentences were presented in the center of the screen in 30 pt size, at a rate of 400 ms per word. Participants saw a blank screen for 100 ms between each word. Participants were asked to press the P key to indicate that a sentence is acceptable and Q to indicate unacceptability. They were instructed to provide judgments as quickly as possible. A warning message in red font appeared if they did not respond within 10 seconds.

14.6.4 Analysis

Statistical analysis was carried out by fitting a Bayesian hierarchical Bernoulli model to *yes* responses given to experimental items. We used Stan (Stan Development Team 2020) and the brms package in R (Bürkner 2017), with weakly informative priors, maximal random effects, and sum-coded predictors. The model-fitting specifications used in brms are reported in Table 14.1. The contrasts of factors are reported in Table 14.2.

Table 14.1 Bayesian model specifications

Intercept prior	<i>Normal</i> (0,1)	SD priors	<i>Normal</i> (0,1)
Slope priors	<i>Normal</i> (0,1)	Correlation priors	<i>LKJ</i> (2)
Formula	yes_responses ~ SA*conjoiner + (SA*conjoiner subject) + (SA*conjoiner item)		

Table 14.2 Contrasts used in the Bayesian model

	+0.5	−0.5
Suspended Affixation	Present	Absent
Conjoiner Type	Non-Harmonizing (<i>ya da</i>)	Harmonizing (<i>ve</i>)

The data for my study, along with the analysis scripts and items, can be found at https://github.com/utkurturk/SA_NanoChapter link.

14.6.5 Results

Figure 14.2 shows the average proportions of ‘yes’ responses in each of the four conditions. The x-axis shows the presence of suspended affixation; the line type shows the conjoiner used. The gray area shows the uncertainty zone. It was calculated by adding the error rate of the grammatical condition to the rate of 50 percent (chance). The graph shows that sentences with harmonizing conjoiners were rated as acceptable as sentences with non-harmonizing conjoiners when the sentences do not have suspended affixation ($M = 0.93$ and 0.93 , $CI = 0.02$ and 0.02 , for vowel matching and mismatching respectively). However, within suspended affixation sentences, participants rated harmonizing conjoiners more acceptable ($M = 0.75$, $CI = 0.03$) than the non-harmonizing conjoiners ($M = 0.67$, $CI = 0.03$). More importantly, the acceptability of sentences with suspended affixation and non-harmonizing conjoiners falls into the uncertainty zone. This is expected and verifies the previous hypothesis that vowel quality mismatch will result in a significant acceptability difference in a suspended affixation context.

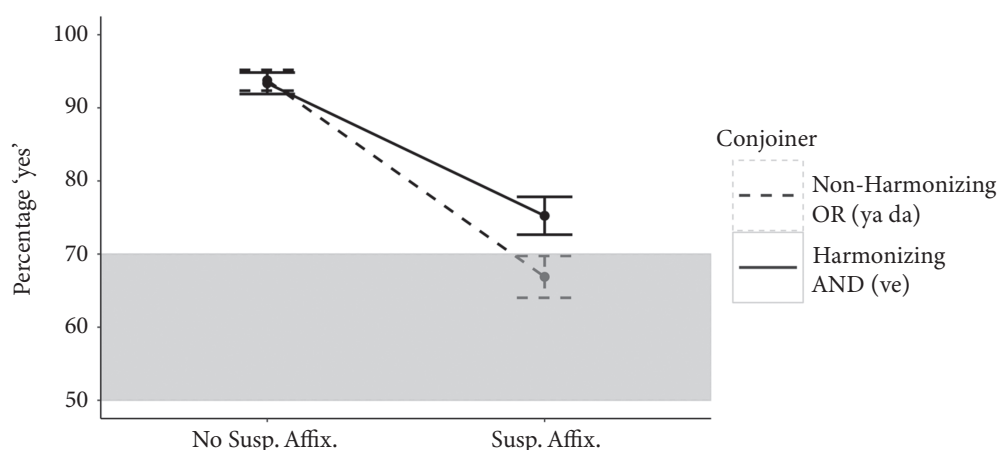


Figure 14.2 The average percentage of acceptable/yes responses according to the experimental conditions in this study. Error bars signal standard errors calculated following Morey (2008)

In Figure 14.3, we see the posterior probabilities for the Bayesian GLM model with a logit link. The negative main effect of conjoiner type ($\beta = -0.29$; $CI = [-0.54; -0.04]$; $P(\beta < 0) > .99$) indicates that, on average, participants gave fewer ‘yes’ responses when the sentence had *ya da* instead of *ve*, as predicted by Schwarz, Clifton Jr, and Frazier (2007). Additionally, the negative main effect of the presence of suspended affixation ($\beta = -2.03$; $CI = [-2.54; -1.53]$; $P(\beta < 0) > .999$) is also significant; that is, participants gave fewer ‘yes’ responses when the first dative marker was dropped, and the form was changed back to the bare form. More important

is the presence of a negative interaction between the conjoiners type and suspended affixation ($\beta = -0.82$; $CI = [-1.32; -0.32]$; $P(\beta < 0) > .999$), meaning that we have strong evidence showing that participants gave fewer ‘yes’ responses to sentences with mismatching conjoiners specifically in the context of suspended affixation.

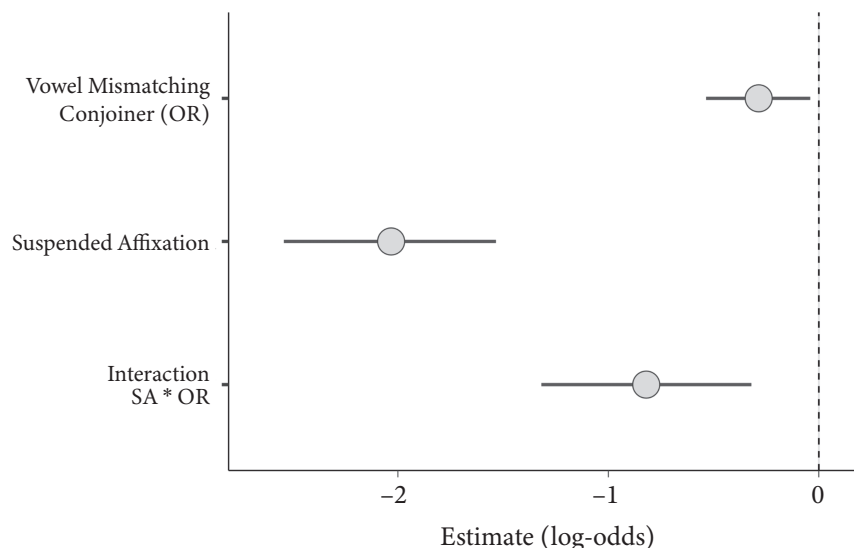


Figure 14.3 Estimates and 89% credible intervals for the logit regression coefficients for the model of responses to experimental trials in the experiment

Table 14.3 shows the mean estimated likelihood and evidence ratio for more than 70 percent acceptance for each condition retrieved from the Bayesian GLM. The important bit of information is that every condition except for the non-harmonizing suspended affixation one has over 99 percent evidence ratio, meaning almost all estimates for every possible sample have over 70 percent likelihood of acceptance. However, this is not the case for non-harmonizing cases in the context of suspended affixation, as expected.

Table 14.3 Estimated likelihood and evidence ratio of over 70%

Condition	Estimated likelihood (%)	Evidence ratio (%)
Suspension + Harmonizing	95	>99
No Suspension + Harmonizing	76	>99
Suspension + Non Harmonizing	94	>99
No Suspension + Non Harmonizing	61	46

14.6.6 Discussion

The experiment showed that there was a significant acceptability difference between *ben=ve* and *ben=ya=da*. Turkish speakers found the presence of a non-vowel-harmonic conjoiner less acceptable in a systematic way. However, the results also show more than chance grammaticality for these items. This increased acceptability might be due to two factors: speaker variability and lack of clearly ungrammatical conditions, biasing people towards saying more ‘yes’ responses (Macmillan and Creelman 2005). Nevertheless, when the general tendency to not accept suspended affixation is accounted for, we see that *ben=ya=da* cases fall into the uncertainty zone.

14.7 Conclusion

In this chapter, I have examined the distribution of personal pronouns in the context of suspended affixation and the behavior of suppletion via the lens of experimental and crosslinguistic data, as well as an original piece of Turkish data. For the analysis of internal structure, distribution of illegitimate suspensions, and suppletion, I have adopted a modified Nanosyntactic model of lexicalization following the proposals of Bye and Svenonius (2012), Svenonius (2012), and Starke (2020).

I propose that suppletive pronouns like *sana* (you.DAT) and *bana* (I.DAT) are not decomposable despite the attractive possibility of *san+a*, which explains their ungrammaticality as a second conjunct under the identity assumption. More importantly, in the context of a conjoiner *ve*, these complex structures are forced to look for an additional lexicalization route, which ends up being the already proposed backtracking procedure for the DAT cases with common nouns in Türk and Caha (2022), which makes them ellipsis-prone syntactic constructions due to having the same DAT structure.

The main contribution of this chapter is that phonological processes may rerank the different candidates for exponence by forcing a reanalysis of a lexicalized structure.

Acknowledgments

I would like to thank Sadira Lewis, Maša Bešlin, and Luisa Seguin for their comments on this version of the chapter. I would also like to extend my thanks to Omar Agha, Furkan Atmaca, Pavel Caha, Furkan Dikmen, Allison Dods, David Erschler, Aron Hirsch, Sebastián Mancha, Kate Mooney, Masha Polinsky, anonymous reviewers of ConSOLE29, and UMD SLab for their comments and contributions

on an early form of this work. Lastly, I would like to thank Balkız Öztürk and Juan Uriagereka for their encouragement and wise cautionary warnings.

Abbreviations

AOR	=	aurist
EVID	=	evidential
IMP	=	imperative
POSS	=	possessive
PRS	=	present
PST	=	past
SPEC	=	specific

References

- Akkuş, Faruk. 2016. "Suspended Affixation with Derivational Suffixes and Lexical Integrity." In *Mediterranean Morphology Meetings*, ed. Jenny Audring, Francesca Masini, and Wendy Sandler. Haifa, Israel: University of Haifa, 10: 1–15.
- Atmaca, Furkan. 2022. "Suspended Affixation Needs No Morphological Word: The Suffix-(y)Ip." In *Proceedings of the Workshop on Turkic and Languages in Contact with Turkic*, ed. Songül Gündoğdu, Sahar Taghipour, and Andrew Peters. Linguistic Society of America, 6: 5035.
- Béjar, Susana. 2003. "Phi-Syntax: A Theory of Agreement." PhD dissertation, University of Toronto.
- Bozşahin, Cem. 2007. "Lexical Integrity and Type Dependence of Language." MS, METU.
- Broadwell, George. 2008. "Turkish Suspended Affixation Is Lexical Sharing." In *Proceedings of the LFG08 Conference*, ed. Miriam Butt and Tracy Holloway King. Center for the Study of Language and Information, 1–16.
- Bürkner, Paul-Christian. 2017. "Brms: An R Package for Bayesian Multilevel Models Using Stan." *Journal of Statistical Software* 80(1): 1–28.
- Bye, Patrik and Peter Svenonius. 2012. "Non-Concatenative Morphology as Epiphenomenon." In *The Morphology and Phonology of Exponence*, ed. Jochen Trommer. Oxford: Oxford University Press, 427–95.
- Caha, Pavel. 2009. "The Nanosyntax of Case." PhD dissertation, Universitetet i Tromsø.
- Caha, Pavel. 2019. "Case Competition in Nanosyntax: A Study of Numerals in Ossetic and Russian." <https://ling.auf.net/lingbuzz/004875> (accessed December 1, 2019).
- Drummond, Alex. 2013. "Ibex Farm." <https://spellout.net/ibexfarm> (accessed February 9, 2021).
- Erschler, David. 2012. "Suspended Affixation in Ossetic and the Structure of the Syntax–Morphology Interface." *Acta Linguistica Hungarica* 59(1–2): 153–75.
- Erschler, David. 2018. "Suspended Affixation as Morpheme Ellipsis: Evidence from Ossetic Alternative Questions." *Glossa: A Journal of General Linguistics* 3(1): 1–30.
- Göksel, Aslı and Celia Kerslake. 2004. *Turkish: A Comprehensive Grammar*. Routledge.
- Gong, Zhiyu Mia. 2021. "Postsyntactic Lowering and Linear Relations in Dagur Noun Phrases." *Glossa: A Journal of General Linguistics* 6(1): 1–29.

- Gračanin-Yüksek, Martina. 2016. "Alternative Questions in Turkish." *Dilbilim Araştırmaları Dergisi* 1: 1–13.
- Gruzdeva, Ekaterina. 1998. *Nivkh*. Munich: Lincom Europa.
- Guseva, Elina and Philipp Weisser. 2018. "Postsyntactic Reordering in the Mari Nominal Domain." *Natural Language & Linguistic Theory* 36(4): 1089–127.
- Kabak, Barış. 2007. "Turkish Suspended Affixation." *Linguistics* 45(2): 311–47.
- Kornfilt, Jaklin. 1996. "On Copular Clitic Forms in Turkish." *ZAS Papers in Linguistics* 6: 96–114.
- Kornfilt, Jaklin. 2012. "Revisiting 'Suspended Affixation' and Other Coordinate Mysteries." In *Functional Heads, Volume 7: The Cartography of Syntactic Structures*, ed. Laura Brugé, Anna Cardinaletti, Giuliana Giusti, Nicola Munaro, and Cecilia Poletto. Oxford: Oxford University Press, 181–96.
- Macmillan, Neil A. and C. Douglas Creelman. 2005. *Detection Theory: A User's Guide*. Mahwah, NJ: Lawrence Erlbaum Associates.
- Morey, Richard D. 2008. "Confidence Intervals from Normalized Data: A Correction to Cousineau (2005)." *Reason* 4(2): 61–4.
- Schwarz, Florian, Charles Clifton Jr, and Lyn Frazier. 2007. "Strengthening 'or': Effects of Focus and Downward Entailing Contexts on Scalar Implicatures." *University of Massachusetts Occasional Papers in Linguistics* 33(1): 9.
- Stan Development Team. 2020. "RStan: The R Interface to Stan."
- Starke, Michal. 2009. "Nanosyntax: A Short Primer to a New Approach to Language." *Nordlyd* 36(1): 1–6.
- Starke, Michal. 2013. "Nanosyntax, Part II." Lecture series at CRISSP, Brussels.
- Starke, Michal. 2017. "Resolving (DAT = ACC) ≠ GEN." *Glossa: A Journal of General Linguistics* 2(1): 1–8.
- Starke, Michal. 2018. "Complex Left Branches, Spellout, and Prefixes." In *Exploring Nanosyntax*, ed. Lena Baunaz, Liliane Haegeman, Karen De Clercq, and Eric Lander. Oxford: Oxford University Press, 239–49.
- Starke, Michal. 2020. "Nanoseminar." Online lecture series at Masaryk University, Brno.
- Svenonius, Peter. 2012. "Lexical Insertion in Two Stages." <https://ling.auf.net/lingbuzz/001580> (accessed May 20, 2023).
- Trommer, Jochen. 2008. "Case Suffixes, Postpositions, and the Phonological Word in Hungarian." *Linguistics* 46(2): 1–26.
- Türk, Utku and Pavel Caha. 2022. "Nanosyntactic Analysis of Turkish Case System." In *Proceedings of the Workshop on Turkic and Languages in Contact with Turkic*, ed. Songül Gündoğdu, Sahar Taghipour, and Andrew Peters. Linguistic Society of America, 6: 5051.
- Wyngaerd, Guido Vanden. 2018. "The Feature Structure of Pronouns." In *Exploring Nanosyntax*, ed. Lena Baunaz, Liliane Haegeman, Karen De Clercq, and Eric Lander. Oxford: Oxford University Press, 277–304.
- Yoon, James Hye Suk and Wooseung Lee. 2005. "Conjunction Reduction and Its Consequences for Noun Phrase Morphosyntax in Korean." In *Proceedings of the 24th West Coast Conference on Formal Linguistics*, ed. John Alderete, Chung-hye Han, and Alexei Kochetov. Somerville, MA: Cascadilla Proceedings Project, 379–87.